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EDUCATION

Shandong University, Qingdao

2024 – 2028

Institute of Frontier and Interdisciplinary Science · class of 2024, expected 2028

Coursework: theoretical mechanics, electrodynamics, thermodynamics and statistical physics, quantum mechanics, advanced quantum mechanics, quantum field theory, particle physics, general relativity.

RESEARCH INTERESTS

Light-front physics: light-front quantization, the light-cone Fock-basis expansion of baryons.
Proton structure: collinear PDFs, transverse-momentum distributions and 3D parton distributions, constrained jointly by DIS, SIDIS and Drell–Yan.

RESEARCH DIRECTIONS & WORK IN PROGRESS

- **Light-front quantization (PH 01).** Kinematics turn multiplicative and the vacuum gets simple, which pushes all the difficulty into the Fock-space truncation. I work on where that truncation can be trusted.
- **Baryon light-cone Fock expansion (PH 02).** Wave functions expanded in the light-cone Fock basis, three-quark sector first; convergence, and which observables survive a cut at fixed N.
- **Proton structure — DIS / SIDIS / DY (PH 03).** Three processes read together: DIS for the collinear PDF, SIDIS for transverse momentum, DY as the cross-check. The 3D distribution only holds once the three agree.

OPEN QUESTIONS

- Does convergence come from restricting transverse momentum or from the number of Fock components?
- How much do DIS, SIDIS and DY actually disagree on the same $\langle k_T^2 \rangle$?
- In which Q^2 range does a Gaussian width model stop being defensible?

TECHNICAL SKILLS

LANGUAGES	Python, C++, Bash, Git
SYMBOLIC	Mathematica for algebra; numerical integration and fitting
THEORY	Light-front quantization, Fock-basis expansion, PDF and TMD fits, perturbative calculations
TOOLING	LaTeX and TikZ, reproducible notes, Linux and HPC

LANGUAGES

Chinese (native) · English (working proficiency).